
Nathan Xia

Engineering Portfolio

Engineering Science - University of Toronto

Aerospace-focused engineering student working across aeroacoustics, flow control, and computational & experimental fluid dynamics.

Email: nathanxia123@gmail.com | Phone Number: (825) 994-0165 | Location: Toronto, ON

LinkedIn: [My LinkedIn](#)

Website version of this portfolio (recommended): <https://nathanielalt.github.io/portfolio/>

ABOUT ME

My name is Nathan Xia and I am an incoming second year Engineering Science student at the University of Toronto, intending to specialize in Aerospace Engineering. I have a lifelong passion for anything aerospace, and am especially interested in aeroacoustics, flow control, and both experimental and computational fluid dynamics. Beyond aerospace, I also have a strong interest in research, which I have abundantly explored throughout the years. In the future, I hope to continue my studies into graduate school and R&D. I have showcased a few of my notable projects to help supplement my resume and to convince you of my passion!

Throughout these projects, I have gained proficiency in a variety of tools and languages. The most notable ones are listed below:

**University of Calgary Research**

Research Opportunity within the applied geochemistry research group at the University of Calgary.

TOOLS AND LANGUAGES

Fusion 360	SolidWorks	AutodeskCFD
OpenVSP/VSPaero	Python	Arduino IDE
MATLAB	C	Java

PROJECT 01 - UTIAS RESEARCH

Aeroacoustics Parametric Research Project



OVERVIEW

As part of a research internship at the University of Toronto Institute for Aerospace Studies (UTIAS), I worked under Professor Lavoie in the Fluid Control and Experimental Turbulence (FCET) Laboratory. I was selected as one of two first-year students conducting research at UTIAS and independently led a parametric study of remote microphone probe (RMP) calibration for aeroacoustic wind-tunnel measurements.

RMPs provide a convenient alternative to wall-mounted microphones by connecting a remotely located microphone to a surface pressure inlet through a tube. However, propagation through the tube attenuates the pressure signal and introduces a phase shift, requiring the probe to be calibrated using a transfer function (TF). These transfer functions are sensitive to geometric and environmental conditions, which can introduce considerable variability between calibrations and affect experimental results. My study investigated 16 variables, including tube length, tube diameter, temperature, and humidity, to better understand their influence and identify configurations that could improve calibration consistency.

I was responsible for the full experimental workflow, from apparatus and test-matrix design through manufacturing, data acquisition, signal processing, statistical analysis, and presentation of the results at the University of Toronto Undergraduate Engineering Research Day (UnERD). I developed a MATLAB acquisition and analysis pipeline that used FFT-based processing to calculate and compare transfer functions across test conditions.

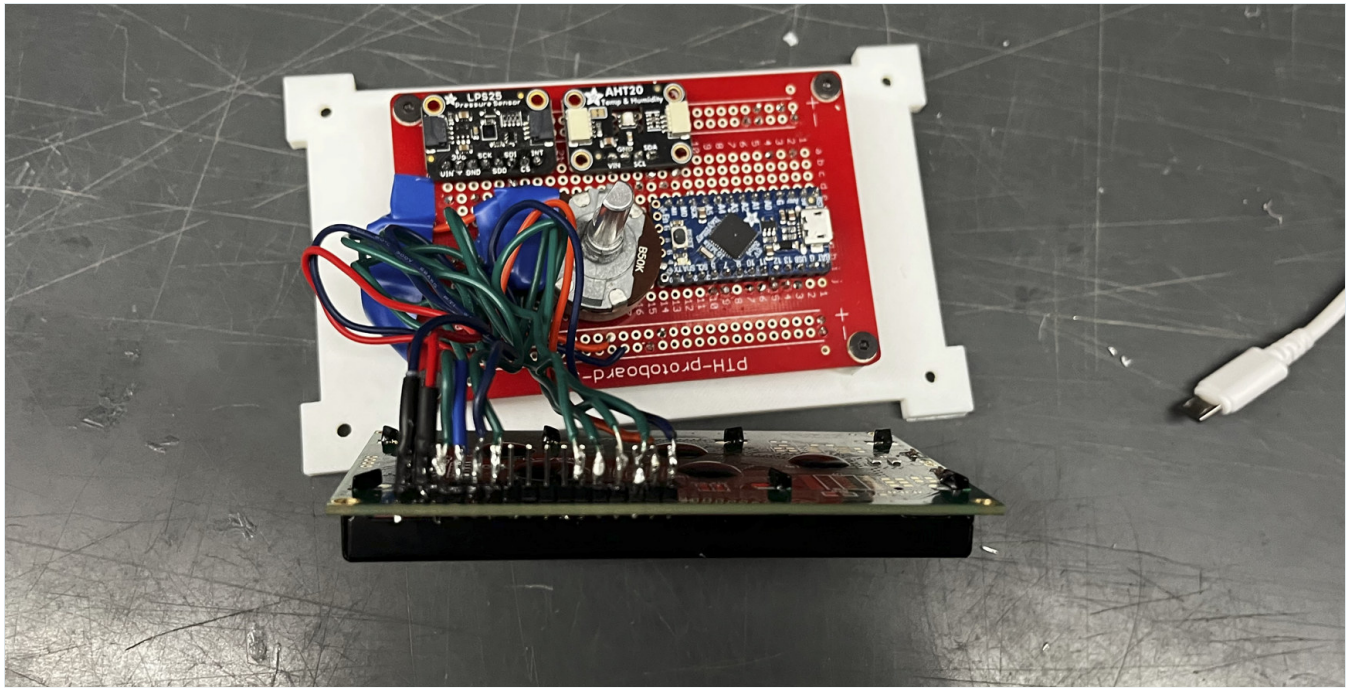
Interpreting the results required relating features in the transfer functions, including attenuation, phase shifts, resonant peaks, and other spectral artifacts, to the underlying acoustic and fluid behaviour within the pneumatic tubing. I compared these observations with analytical models of wave propagation, tube resonance, acoustic impedance, and viscothermal attenuation. This included examining how near-wall boundary layers and other tube-flow effects contributed to frequency-dependent pressure losses.

To support the study, I designed and manufactured parts using SLA printing. Furthermore, I revised the calibration apparatus used by the group and developed a sensor circuit incorporating I2C and SPI communication protocols. Both the apparatus and sensor circuit were retained by the FCET research group for use in subsequent calibration work. The study also identified several geometric and procedural improvements that I proposed to the group, helping inform its continued RMP research and future aeroacoustic measurement practices. Together, the experimental findings, revised apparatus, sensor circuit, and analysis pipeline provided the group with a more repeatable foundation for future probe calibration and testing.

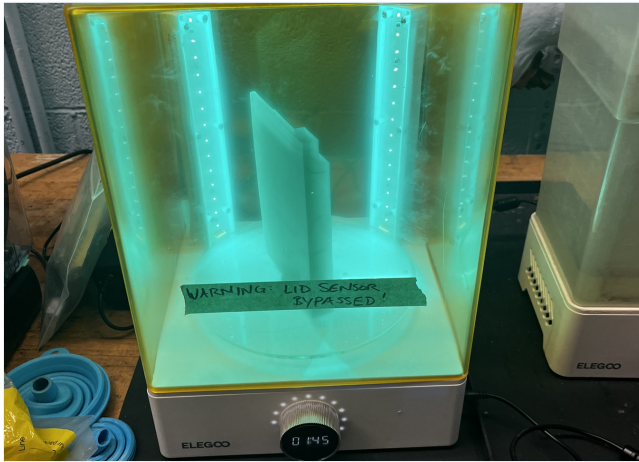
TOOLS USED

Fusion 360	Arduino IDE	Soldering, Circuit Design
SLA and FDM Printing / Slicing	Python & MATLAB (signal processing, algorithms)	DAQ & data collection
Signal Processing and Analysis		

PICTURES

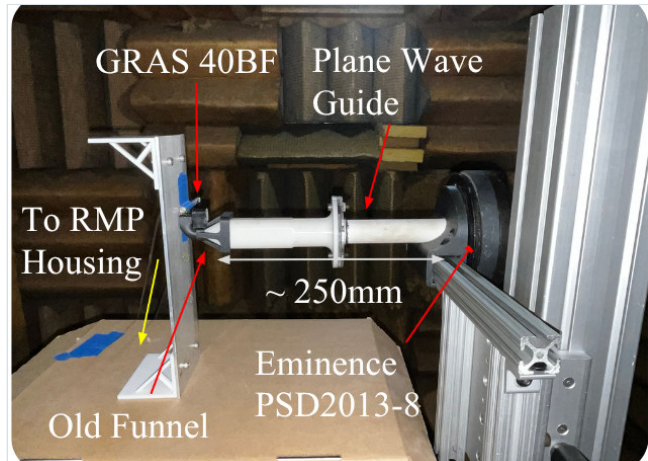
**Circuit Internals**

Completed sensor circuit box, with a working LCD display. Used to read the ambient temperature, humidity, and pressure in the room. It utilizes two sensors, one using I2C protocol and the other using SPI protocol. Communicates through an Adafruit ItsyBitsy microcontroller programmed in Arduino IDE. All data is sent to a SPI LCD display, where brightness is tuned using a built-in potentiometer. Calibration was done manually to verified sources.



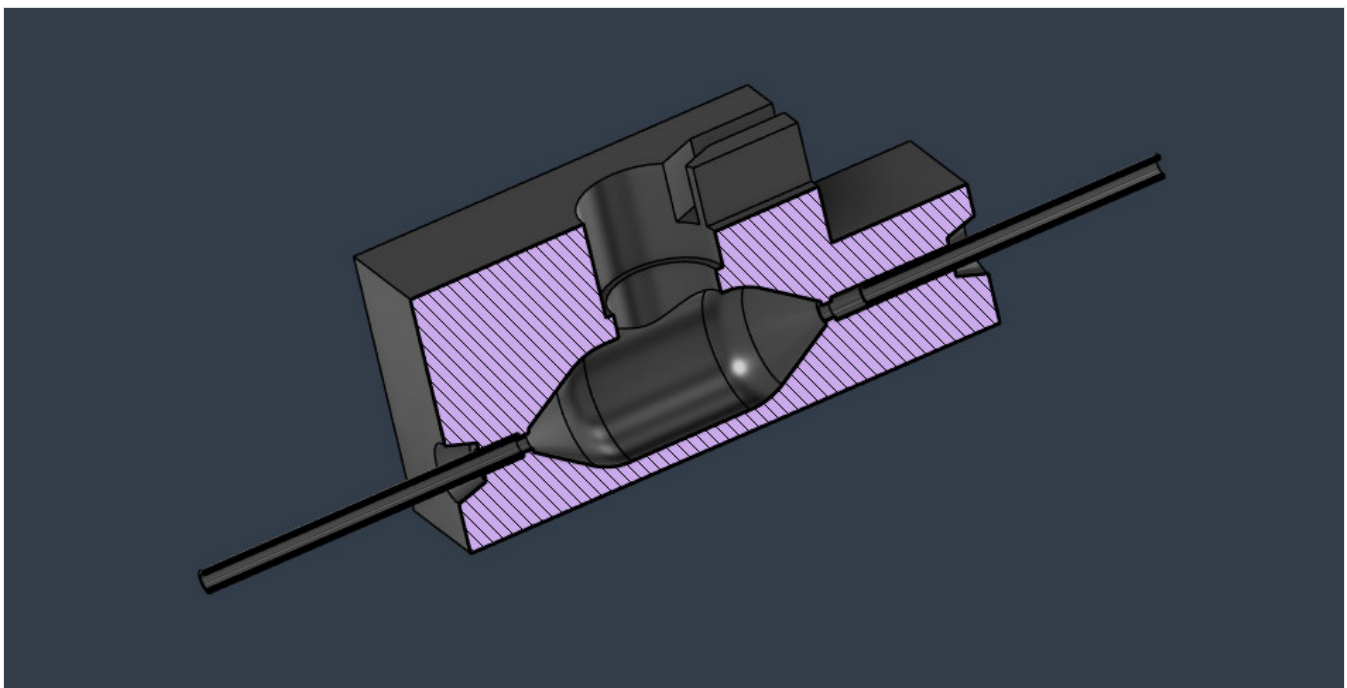
Resin Curing

SLA printing and curing using water-washable resin. Resin was used over FDM printers because of its small layer lines. These small layers guarantee that there is no acoustic leakage in the tubes printed within. All pores and openings were subsequently coated in epoxy to prevent leakage.



Testing Apparatus

The full and labelled testing apparatus in an anechoic chamber used to collect data for this study. All data was collected using a DAQ at a sampling rate of 2^{16} , with an FFT window size of 2^{12} . Further data collection and analysis was done using MATLAB. The full script can be found here: <https://github.com/nathanielalt/UTIAS-2026-MATLAB-Script>



RMP CAD

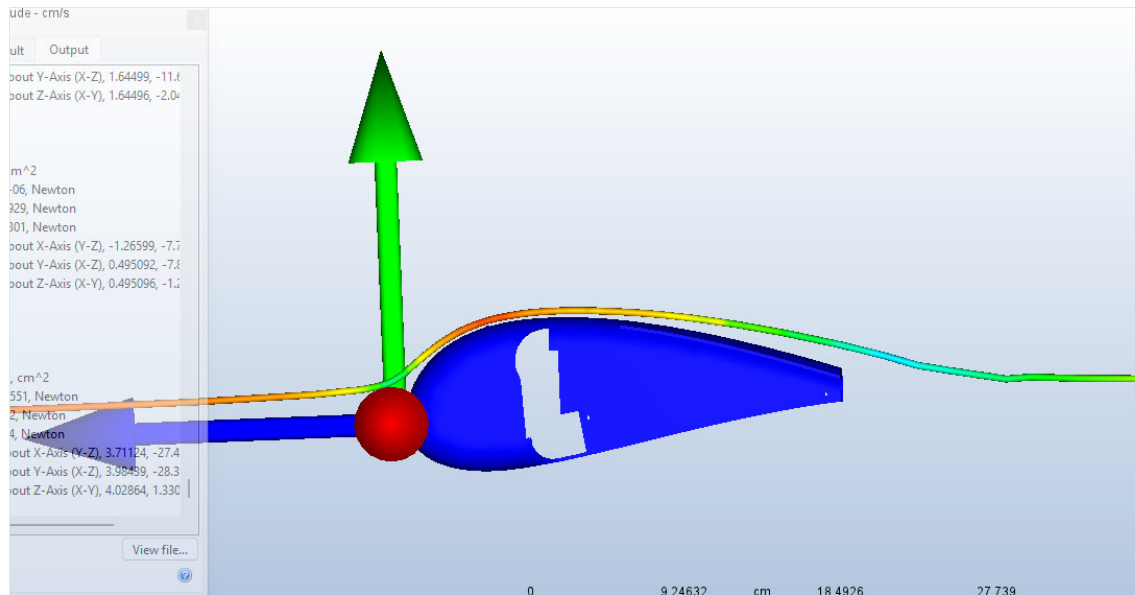
CAD of the RMP Housing used to contain the microphone in this research study. CAD was done by taking measurements from an existing print and redesigning the part in fusion to be manipulated.

RESULTS - Effectively outlined the relationship between various parameters and RMP transfer functions, significantly increasing the groups understanding - Extracted and experimentally validated RMP improvements informed by insights revealed by the dataset - Determined an improved RMP configuration, improving repeatability by 69%, decreasing tortuosity by 46%, and increasing coherence by 4.6% at frequencies below 300 Hz. - Delivered valuable data insight and tools (such as the sensor circuit, redesigned calibration apparatus) to the team at FCET to further optimize future RMP configurations. - Presented findings at U of T's UnERD research conference	CHALLENGES - Sensor Calibration: Ensuring that the sensor housed inside the container accurately reflected the ambient conditions of the room rather than its internal microbiome - Experimental Design: Isolating over 16 variables to ensure accurate data between manipulations - Isolating and converging on the source of variability - Spotting patterns within experimental and mapping those results to literature - Determining methods to mathematically quantify improvements in the TF - Manufacturing: Designing and slicing prints resistant to SLA warping and support failure
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PROJECT 02 - RSX AERIAL DESIGN

Cansat Autonomous Glider



OVERVIEW

As part of the RSX Aerial Design Team, I was primarily responsible for the aerodynamic development of an autonomous glider for the AAS CanSat Competition within the mechanical subsystem. I proposed the lifting-body fuselage concept that was ultimately selected for the final vehicle and led much of its aerodynamic development, from early configuration studies through analysis and design refinement.

I developed a staged workflow that combined rapid, lower-fidelity analysis with more detailed CFD. I first used OpenVSP and VSPAERO to compare more than 20 candidate airfoil profiles and explore a range of fuselage geometries. The most promising configurations were then selected for higher-fidelity analysis in Autodesk CFD, with key results and flow trends also cross-checked using ANSYS Fluent.

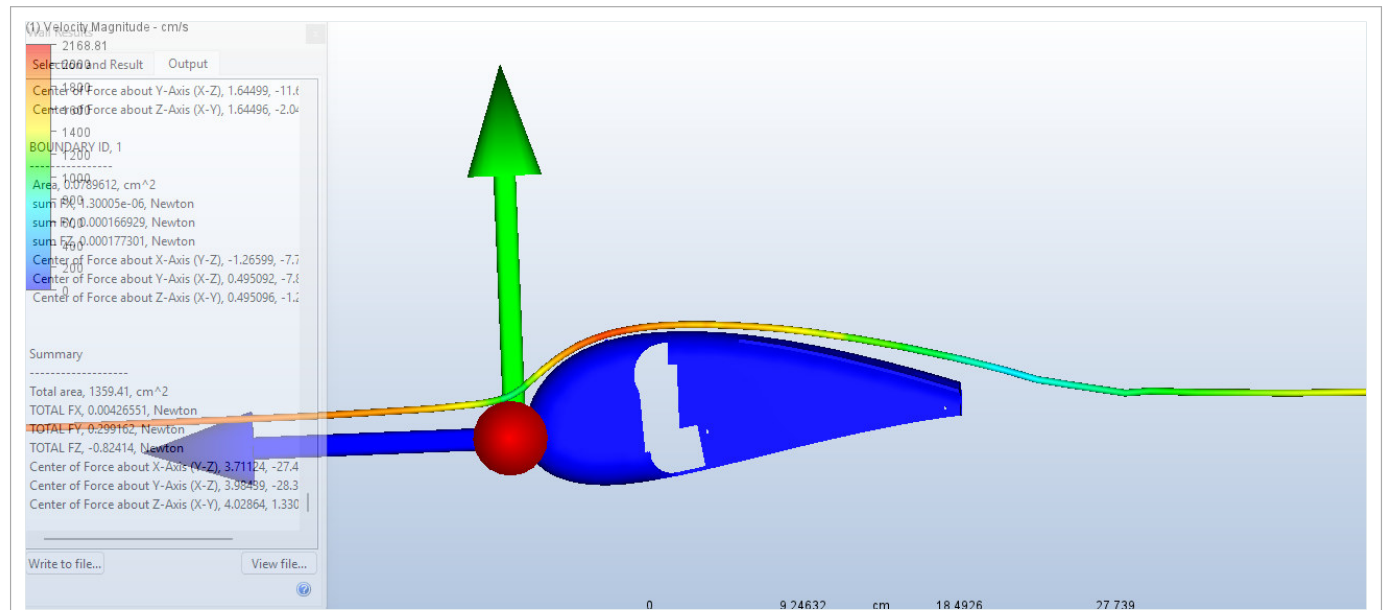
The CFD results helped me examine pressure distributions, streamlines, wake development, and boundary-layer behaviour around the lifting-body fuselage. This analysis indicated premature flow separation and considerable spanwise and tip-flow losses, particularly at the glider's low-Reynolds-number operating conditions. Based on the observed flow behaviour, I incorporated vortex generators and endplates as passive flow-control devices. In simulation, these changes significantly delayed separation and produced an approximately fivefold increase in lift at the design condition.

Beyond the glider fuselage, I evaluated rocket nose-cone geometries using Autodesk CFD and OpenRocket, supported CFD-based verification of control mechanisms and aerodynamic interfaces, and assisted with the fibre-composite manufacture of the flight wings. I also played a major role in the refinement of individual parts for electronic and mechanical integration. A significant portion of the tail and release mechanisms were also manufactured and refined by me.

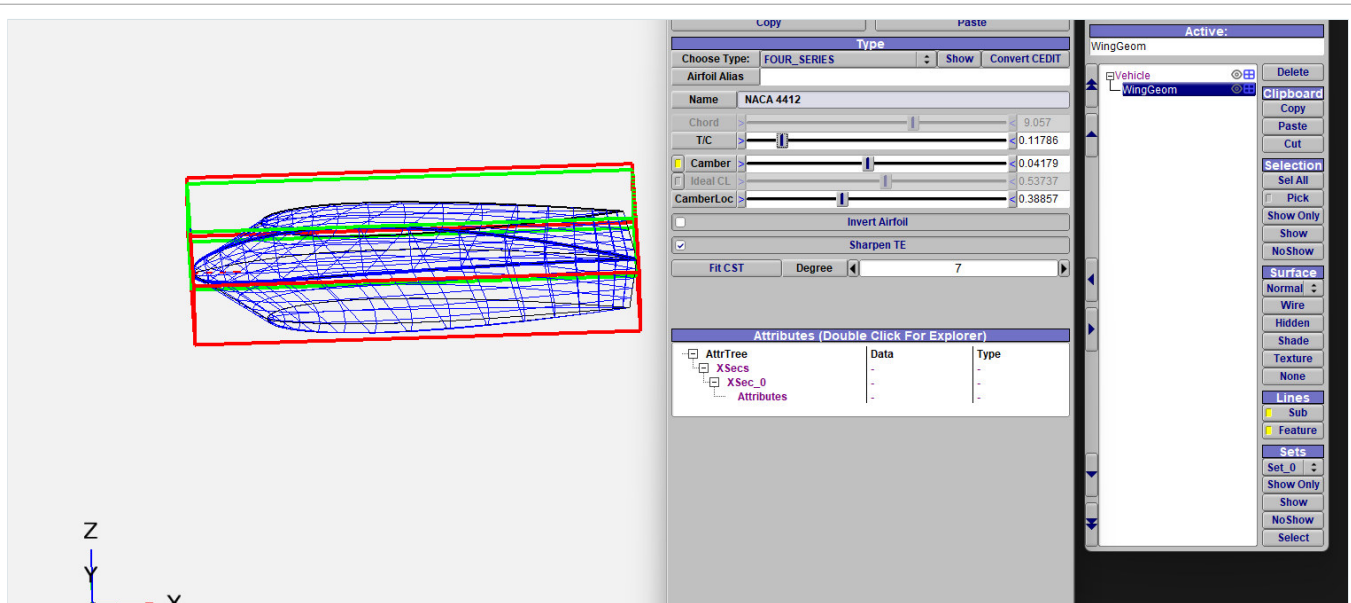
TOOLS USED

Fusion 360	Autodesk CFD	OpenVSP
OpenRocket	OpenVSP	Composite manufacturing
Mechanical and electrical integration		

PICTURES

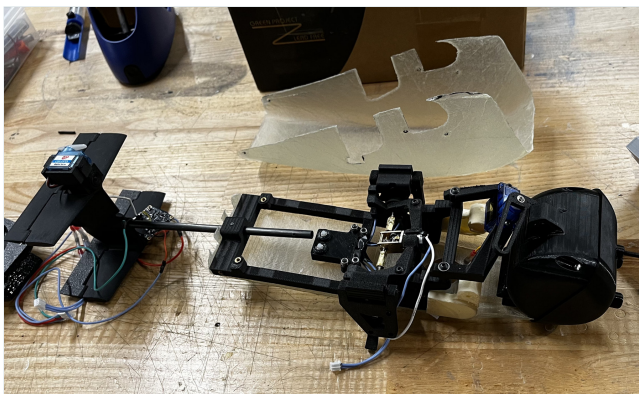
**Glider Body CFD**

CFD showing trace airflow over the optimized glider body. Vortex generators are implemented to mitigate boundary layer separation, and end plates are implemented to prevent side air roll up. A 5x improvement on the lift force was found in comparison to the raw design, along with a significant delay in boundary layer separation.



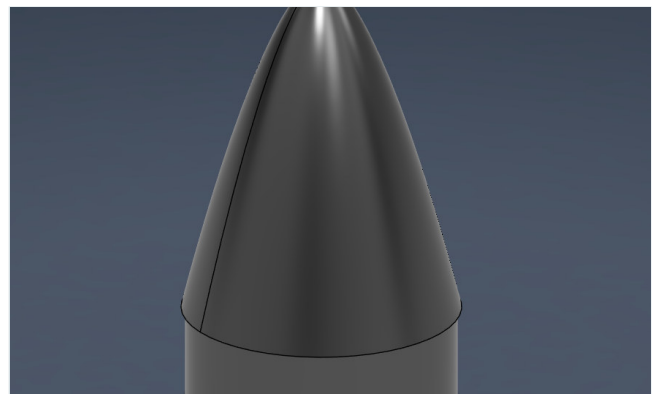
VSPAero

Example of an airfoil profile tested in VSPAero. All initial simulations were done in this application at an Re of ~ 97000 . The resulting Cl graphs as a function of AOA were compared between each geometry, and the most promising profiles were taken for further validation. These designs being the NACA 4421, NACA 4424, and the fx69pr281.



T-Tail Manufacturing

Final glider design, featuring the glider fuselage that I proposed and designed. The glider frame was 3D printed using a nylon/carbon fibre infused composite. The wings and glider body were both made out of a fibre-glass/resin composite which I helped manufacture. A significant portion of the tail was also verified by me in CFD and refined to aid integration.



CANSAT Nosecone

CAD on the payload nosecone, implementing a power-series geometry profile to minimize drag. Numerous geometries were first analyzed using a low-fidelity software (OpenRocket). Promising designs were then 3D modelled and validated using Autodesk CFD.

RESULTS

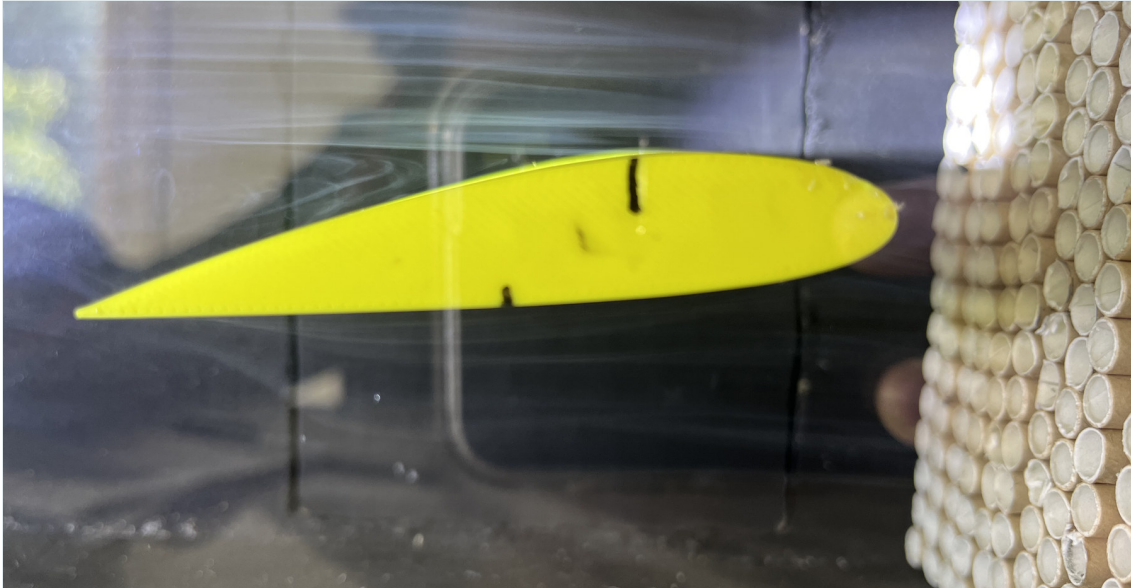
- Competition results scored 70% overall
- Placed 2nd in Canada
- Placed 8th in North America

CHALLENGES

- Integration and manufacturing of mechanical components
- Reconciling CAD models to manufacturing limitations
- Working under time pressure and with a large design team

PROJECT 03 - SCIENCE FAIR

Flow Control Research Project



OVERVIEW

As part of an independent passion project that later developed into a national science fair entry, I investigated how vortex generator placement and long-gap spacing affected boundary-layer separation on a NACA 2412 airfoil. The profile was selected for its relevance to beginner aircraft such as the Cessna 172. This project was my first experience conducting aerodynamic research and experimentally evaluating passive flow-control devices.

I designed and built a low-speed flow-visualization wind tunnel at home using limited materials, with much of the structure constructed from cardboard and hot glue. The tunnel operated at approximately 0.84 m/s and a Reynolds number below 2,300. Without access to formal laboratory facilities or dedicated aerospace mentorship, I developed the experimental apparatus, test procedure, and analysis approach using the resources available to me.

I mathematically generated and 3D printed eight 75 mm chord NACA 2412 airfoils, each incorporating a different vortex-generator configuration. I developed a controlled test matrix that compared four chordwise locations, from 10% to 25% chord, and four long-gap spacings, from 2 mm to 5 mm. Smoke visualization was used to observe boundary-layer behaviour as the angle of attack increased, with separation reaching 30% chord used as a consistent stall-proximity criterion. Each configuration was tested across three trials to compare its ability to delay separation.

The results indicated that vortex-generator location had a greater effect on separation delay than long-gap spacing. Parabolic regression of the experimental data identified predicted optima of 17.2% chord for placement and 3.6 mm for long-gap spacing, with coefficients of determination above 0.936. The optimized placement increased the predicted separation angle from 6.7 degrees for the control airfoil to 11.8 degrees, representing an approximately 76% improvement. The testing also showed that poorly selected configurations could perform worse than the untreated airfoil, suggesting that vortex generators require deliberate optimization rather than simply being added in greater numbers.

Despite the low-cost apparatus and limited resources, the project produced clear experimental trends and progressed from a personal investigation to the Canada-Wide Science Fair (CWSF). It received a Gold Medal and the \$500 Second Best Senior Project Award at the Calgary Youth Science Fair. It was then selected as one of 15 projects from approximately 600 entries to advance to CWSF, where it received a Bronze Medal among approximately 350 competing projects.

TOOLS USED

CAD	Desmos - Airfoil Design	Experimental design, analysis & communication
Flow Control	Wind Tunnel testing and analysis	

PICTURES

**Airflow in Wind Tunnel**

Flow Separation visualized on a NACA2412 airfoil in the wind tunnel. A recirculation region can be seen. Incense smoke was used to visualize the airflow over the wing. Airfoil was modeled in Desmos before converting into a .STL file in TinkerCAD. This was my first experience with CAD.



Wind Tunnel Fan

After the suction from an electronic fan was insufficient, a 3D Modeled fan was used to generate enough suction force inside the wind tunnel. This fan was inspired by the suction fans used inside of vacuum cleaners.

Low Speed Visualization Wind Tunnel

Lighter

Backplate

Incense

Wooden Dowels

Incense Stand

Hot glue gun

NACA 2412 Airfoils

Flashlight

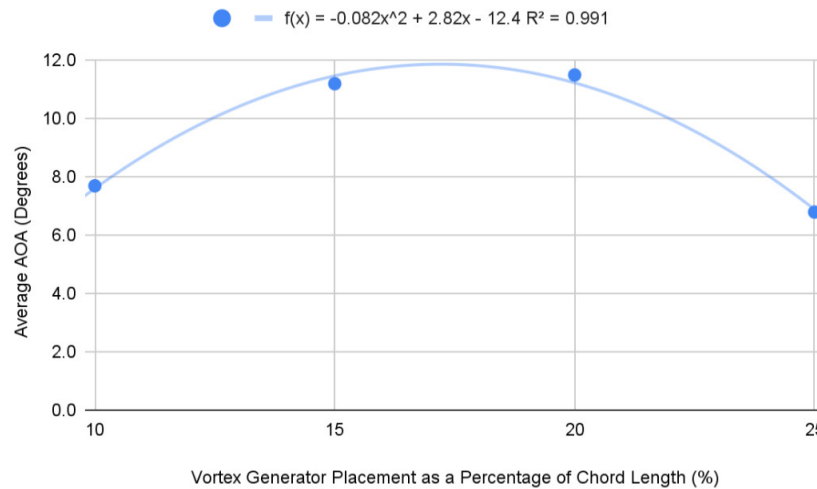
Sharpie



Labelled Apparatus

Complete and labelled materials and apparatus diagram used to collect airfoil data. Eight different airfoils with vortex generators at varying locations and spacings can be seen. The home-made wind tunnel is also pictured.

Average Boundary Layer Separation Angle as a Function of Vortex Generator Placement



Experimental Data Plot

Example data plot acquired from the wind tunnel apparatus. A clear parabolic trend can be seen with the data plots, with a distinct peak in the middle.

RESULTS

- Placed 2nd Overall for seniors at the Calgary Youth Science Fair (CYSF)
 - \$500 Cash Prize
- 1 of 15 teams out of ~600 projects displayed selected to advance to the Canada-Wide Science Fair (CWSF)
- Awarded a bronze medal at the CWSF, out of a total of ~350 projects
- Effectively demonstrated the impact of passive flow control techniques
- Effectively visualized the flow around an airfoil at different AOA, given limited resources and knowledge

CHALLENGES

- Figuring out how and where to begin, with limited resources and experience
- Designing and Manufacturing a home-made wind tunnel
 - Designing and Manufacturing a home-made wind tunnel
 - Visualizing airflow within wind tunnel
 - Generating enough suction within the wind tunnel
 - Designing an effective data acquisition method to measure stall AOA

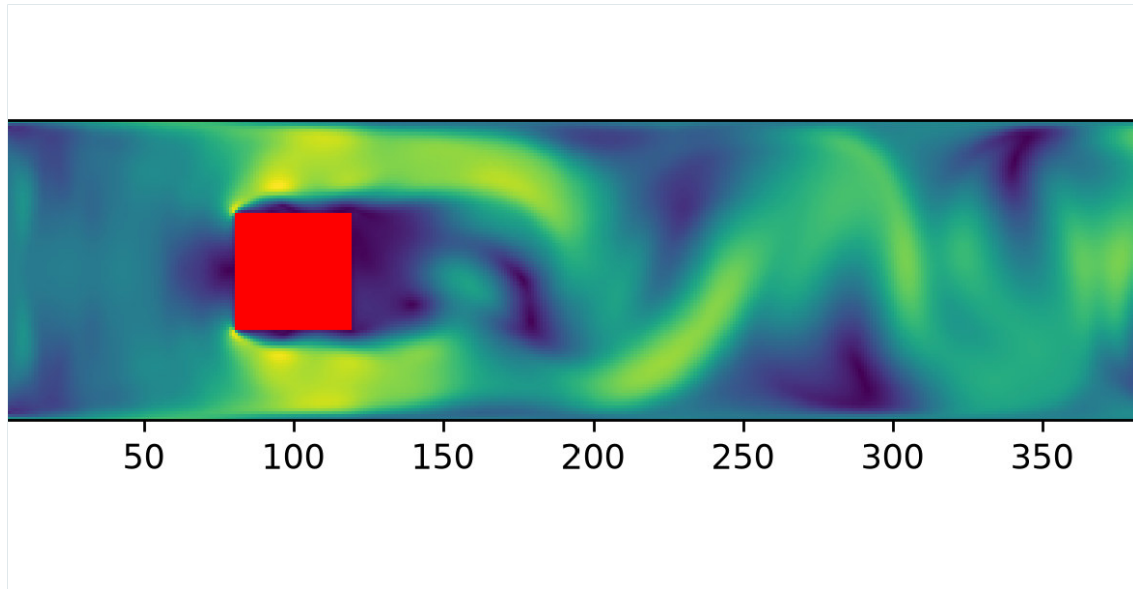
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PROJECT 04 - PERSONAL PROJECT

Beginner's CFD Ongoing



OVERVIEW

This project is still ongoing. Many current day CFD software such as AutodeskCFD or Fluent require significant effort to simply run a simulation.

Much of the headache is attributed to downloading the software itself and setting up the environment. However, for the majority of beginners, hobbyists and even design teams, much of the advanced settings are unused and only serve to clutter the UI.

This project serves to solve this issue by making a simplified beginner friendly CFD to remove the barrier of entry. Photos, results and challenges will be updated as the project continues.

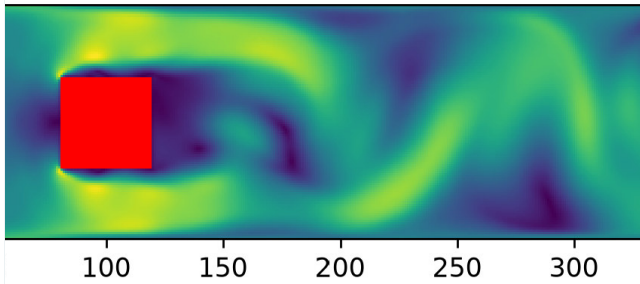
Update 1: Stability is reached for $\tau = 0.51$, collision operator is refined, LES simulation is implemented

Update 2: LES simulation is fixed, refined inlet and outlet conditions, as well as boundary conditions

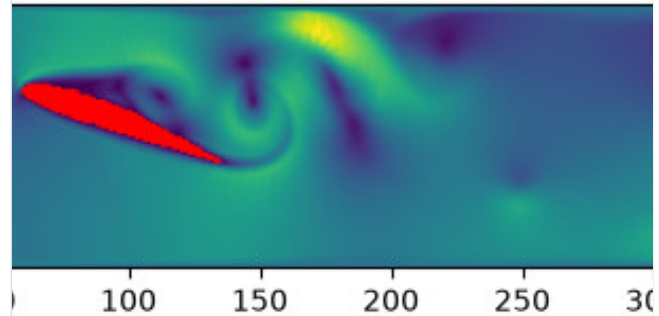
Update 3: Forces can be extracted, verification can begin

Update 4: Strouhal number verification for various shapes is completed.

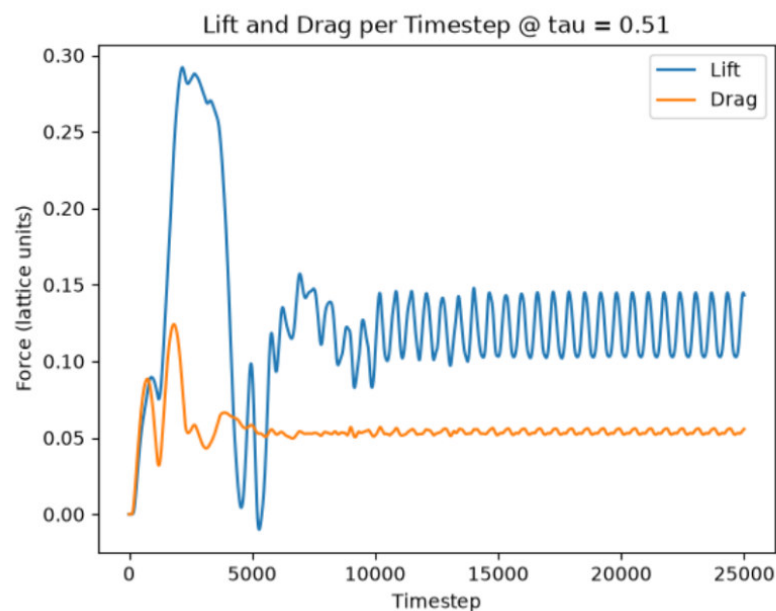
PICTURES

**LBM Simulation**

Flow simulated around a square geometry. Velocity is plotted with low velocities being blue and faster velocities being green/yellow. Notable Von Karmann vortices can be seen behind the cube. Stability is maintained even around the square corners.

**Circle Simulation Curl**

Flow velocity visualization around a NACA 2412 airfoil at 20 degrees AOA. Visible flow separation can be seen, as well as periodic trailing edge vortex shedding.

**Simulated Lift and Drag**

Strouhal number verification for a NACA 0012 airfoil at 8 deg AOA and a Reynolds number of 2000. 25000 timesteps are plotted. The lift oscillations reveal well-behaved periodic vortex shedding beyond ~12000 time steps. Extracting this data corresponds to a chord-wise Strouhal number of 1.15 ± 0.012 , which falls within 1.7% of an accepted and published Strouhal number (1.17) for the same geometry and conditions. My LBM solver produced a value within 1.7% of a published value simulated using ANSYS Fluent.